

Explanatory Analysis

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Getting in The Data and Wrangling

```
# load in necessary packages  
library("tidyverse")
```

```
-- Attaching core tidyverse packages ----- tidyverse 2.0.0 --  
v dplyr      1.1.4      v readr      2.1.5  
v forcats    1.0.0      v stringr    1.5.1  
v ggplot2     4.0.2      v tibble     3.2.1  
v lubridate   1.9.4      v tidyr      1.3.1  
v purrr       1.0.4  
-- Conflicts ----- tidyverse_conflicts() --  
x dplyr::filter() masks stats::filter()  
x dplyr::lag()     masks stats::lag()  
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become
```

```
library("brms")
```

```
Loading required package: Rcpp  
Loading 'brms' package (version 2.23.0). Useful instructions  
can be found by typing help('brms'). A more detailed introduction  
to the package is available through vignette('brms_overview').
```

```
Attaching package: 'brms'
```

```
The following object is masked from 'package:stats':
```

```
ar
```

```
library("ggeffects")
library("readxl")
library("RColorBrewer")
```

```
# read in the data from the excel file
Poll_Data <- read_xlsx("Ellie_ChiSquare.xlsx", sheet = 1)
```

Warning: Expecting numeric in A1192 / R1192C1: got 'male'

```
# make sure the data file is in correctly
head(Poll_Data)
```

```
# A tibble: 6 x 28
  index UniqueID DateCollected      Site Collector Method PollinatorOrder
  <dbl>   <dbl>   <dtm>          <chr>   <chr>    <chr>   <chr>
1    876     956 2021-07-11 00:00:00 Chadwick Jenny Hsu   Netti~ Hymenoptera
2    893    1021 2021-07-11 00:00:00 Chadwick Gabriella ~ Netti~ Hymenoptera
3    892    1020 2021-07-11 00:00:00 Chadwick Gabriella ~ Netti~ Hymenoptera
4    902    1030 2021-07-11 00:00:00 Chadwick Gabriella ~ Netti~ Hymenoptera
5    900    1028 2021-07-11 00:00:00 Chadwick Gabriella ~ Netti~ Hymenoptera
6    896    1024 2021-07-11 00:00:00 Chadwick Sofie Andr~ Netti~ Hymenoptera
# i 21 more variables: PollinatorGroupGeneral <chr>, PollinatorFamily <chr>,
# PollinatorTribe <chr>, PollinatorGenus <chr>, PollinatorSpecies <chr>,
# PollGenusSpeciesUSE <chr>, PollinatorSex <chr>, FlowerCollectedOn <chr>,
# PlantType <chr>, PlantFamily <chr>, PlantGenus <chr>, PlantSpecies <chr>,
# PlantGenusSpeciesUSE <chr>, Native <chr>, FlowerColor <chr>,
# SingleFlColor <chr>, FlowerShape <chr>, AllPollBeeOnly <chr>,
# PollenNectar <chr>, PinnedBy <chr>, Notes <chr>
```

```
# Create new dataframe for just flower shape
Fl_Shape <- Poll_Data %>%
  # used the select function to keep only 4 columns
  select(PollinatorOrder, PollinatorFamily, PollGenusSpeciesUSE, FlowerShape) %>%
  # Get rid of flowers with unknown shape
  filter(FlowerShape != "U")

# remove unnecessary columns to just focus on pollinator order
Fl_Shape_Order <- Fl_Shape %>%
  select(PollinatorOrder, FlowerShape)
```

```
# data frame of 2 flower shapes I want to focus on (removed other flower shapes)
Fl_Shape_Final <- Fl_Shape_Order %>%
  group_by(PollinatorOrder, FlowerShape) %>%
  summarize(n = n()) %>%
  mutate(total = sum(n),
         prop = n / sum(n)) %>%
  ungroup() %>%
  filter(FlowerShape == "Dish" | FlowerShape == "Tube")
```

`summarise()` has grouped output by 'PollinatorOrder'. You can override using the `.groups` argument.

```
# Create new dataframe for just tubular flowers
Tube_Fl_Order <- Fl_Shape_Final %>%
  filter(FlowerShape == "Tube")

# Create new dataframe for just dish flowers
Dish_Fl_Order <- Fl_Shape_Final %>%
  filter(FlowerShape == "Dish")

Dish_Fl_Order
```

```
# A tibble: 3 x 5
  PollinatorOrder FlowerShape      n total  prop
  <chr>           <chr>    <int> <int> <dbl>
1 Diptera         Dish      163   232 0.703
2 Hymenoptera     Dish      563  1118 0.504
3 Lepidoptera     Dish       22    50 0.44
```

Statistical Analysis

```
# Create new dataframe for just tubular flowers
Tube_Fl_Order <- Fl_Shape_Final %>%
  filter(FlowerShape == "Tube")
# Print dataframe to see if it workde correctly
Tube_Fl_Order
```

```
# A tibble: 3 x 5
  PollinatorOrder FlowerShape      n total  prop
```

	<chr>	<chr>	<int>	<int>	<dbl>
1	Diptera	Tube	36	232	0.155
2	Hymenoptera	Tube	421	1118	0.377
3	Lepidoptera	Tube	22	50	0.44

```
m.Tube_Fl_Order <- brm(data = Tube_Fl_Order,
  # Made a beta_binomial distribution and used link = "identity" to not have to back transf
  family = beta_binomial(link = "identity", link_phi = "identity"),
  # categorical model using pollinator order
  n | trials(total) ~ 0 + PollinatorOrder,
  # iterations and warmup alongside number of chains
  iter = 4000, warmup = 2000, chains = 4, cores = 4,
  # setting priors reletively broadly
  prior = c(prior(beta(2, 6), class = "b", dpar = "mu", lb = 0, ub = 1),
    prior(exponential(0.001), class = "phi", lb = 0)),
  file = "output/m.Tube_Fl_Order"
)

# viewing results of model
summary(m.Tube_Fl_Order)
```

```
Family: beta_binomial
Links: mu = identity
Formula: n | trials(total) ~ 0 + PollinatorOrder
Data: Tube_Fl_Order (Number of observations: 3)
Draws: 4 chains, each with iter = 4000; warmup = 2000; thin = 1;
total post-warmup draws = 8000
```

Regression Coefficients:

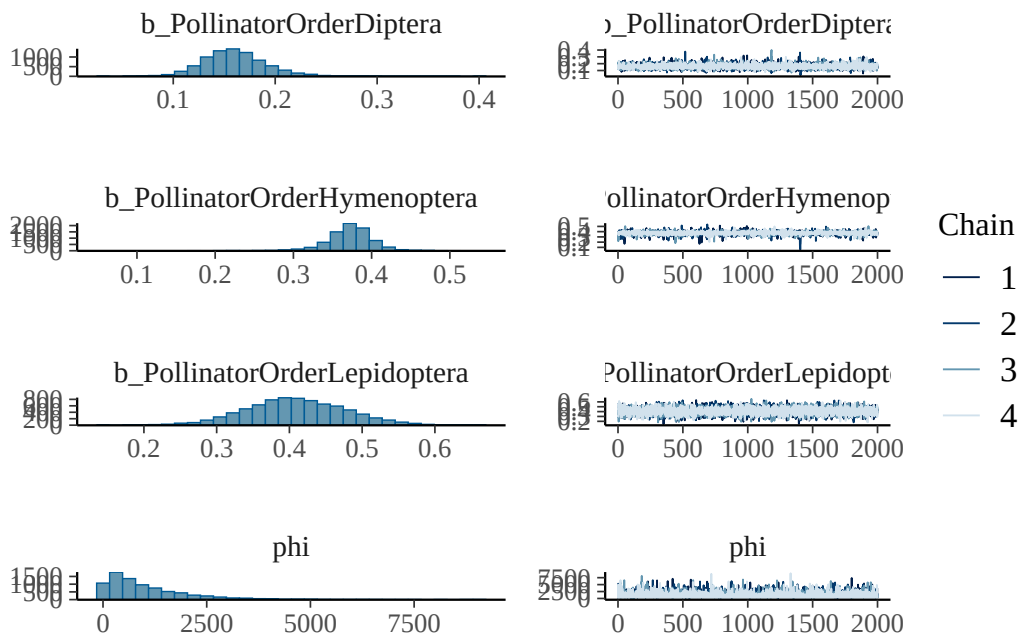
	Estimate	Est.Error	l-95% CI	u-95% CI	Rhat	Bulk_ESS
PollinatorOrderDiptera	0.16	0.03	0.11	0.23	1.00	7038
PollinatorOrderHymenoptera	0.37	0.03	0.30	0.43	1.00	4858
PollinatorOrderLepidoptera	0.41	0.07	0.28	0.54	1.00	7339
	Tail_ESS					
PollinatorOrderDiptera	4868					
PollinatorOrderHymenoptera	2840					
PollinatorOrderLepidoptera	5481					

Further Distributional Parameters:

	Estimate	Est.Error	l-95% CI	u-95% CI	Rhat	Bulk_ESS	Tail_ESS
phi	1043.08	1017.61	38.73	3759.52	1.00	3778	3402

Draws were sampled using sampling(NUTS). For each parameter, Bulk_ESS and Tail_ESS are effective sample size measures, and Rhat is the potential scale reduction factor on split chains (at convergence, Rhat = 1).

```
# show posterior plots and chains to assess model
plot(m.Tube_Fl_Order)
```



```
# Create new dataframe for just dish flowers
Dish_Fl_Order <- Fl_Shape_Final %>%
  filter(FlowerShape == "Dish")
# Print dataframe to see if it workde correctly
Dish_Fl_Order
```

```
# A tibble: 3 x 5
  PollinatorOrder FlowerShape      n total  prop
  <chr>           <chr>      <int> <int> <dbl>
1 Diptera        Dish        163   232 0.703
2 Hymenoptera    Dish        563  1118 0.504
3 Lepidoptera    Dish         22    50 0.44
```

```

m.Dish_Fl_Order <- brm(data = Dish_Fl_Order,
  # Made a beta_binomial distribution and used link = "identity" to not have to back transf
  family = beta_binomial(link = "identity", link_phi = "identity"),
  # categorical model using pollinator order
  n | trials(total) ~ 0 + PollinatorOrder,
  # iterations and warmup alongside number of chains
  iter = 4000, warmup = 2000, chains = 4, cores = 4,
  # setting priors reletively broadly
  prior = c(prior(beta(2, 6), class = "b", dpar = "mu", lb = 0, ub = 1),
    prior(exponential(0.001), class = "phi", lb = 0)),
  file = "output/m.Dish_Fl_Order"
)

# viewing results of model
summary(m.Dish_Fl_Order)

```

```

Family: beta_binomial
Links: mu = identity
Formula: n | trials(total) ~ 0 + PollinatorOrder
Data: Dish_Fl_Order (Number of observations: 3)
Draws: 4 chains, each with iter = 4000; warmup = 2000; thin = 1;
       total post-warmup draws = 8000

```

Regression Coefficients:

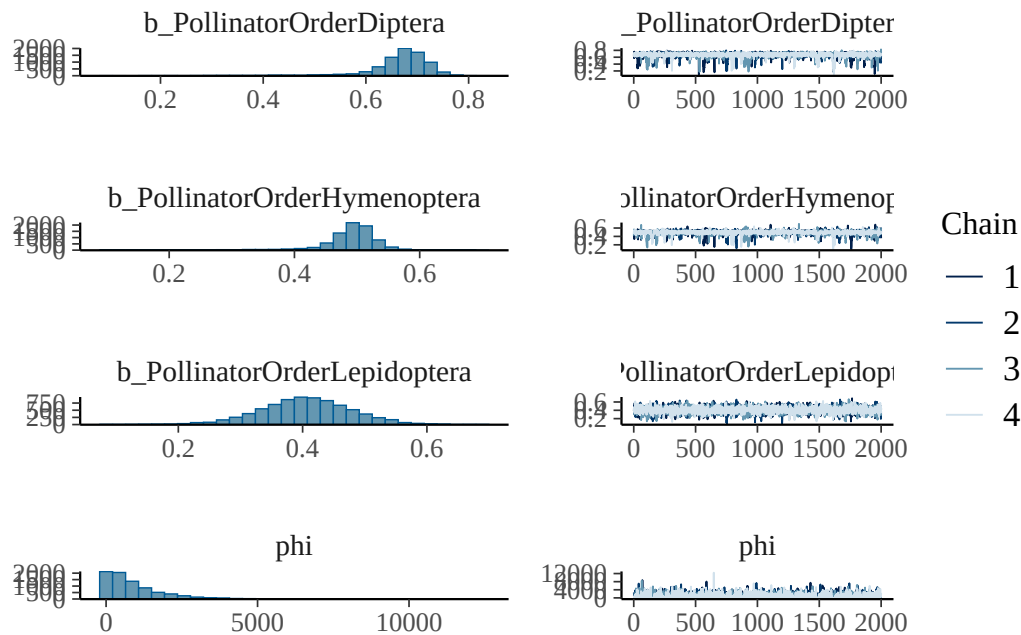
	Estimate	Est.Error	l-95% CI	u-95% CI	Rhat	Bulk_ESS
PollinatorOrderDiptera	0.66	0.07	0.45	0.75	1.00	2253
PollinatorOrderHymenoptera	0.49	0.05	0.33	0.56	1.00	2541
PollinatorOrderLepidoptera	0.40	0.07	0.25	0.55	1.00	3542
	Tail_ESS					
PollinatorOrderDiptera	1025					
PollinatorOrderHymenoptera	1054					
PollinatorOrderLepidoptera	2500					

Further Distributional Parameters:

	Estimate	Est.Error	l-95% CI	u-95% CI	Rhat	Bulk_ESS	Tail_ESS
phi	911.97	994.44	8.37	3612.32	1.00	1430	1049

Draws were sampled using sampling(NUTS). For each parameter, Bulk_ESS and Tail_ESS are effective sample size measures, and Rhat is the potential scale reduction factor on split chains (at convergence, Rhat = 1).

```
# show posterior plots and chains to assess model
plot(m.Dish_Fl_Order)
```



For both of these models we can see that the Rhat values are equal to 1, the posteriors all have one clear hump, and the chains converge. With this we can conclude that the model ran well.

Biologically this means that 16% of Fly visits will be to tubular flowers with a 95% credible interval from [11% - 23%], 37% of Bee visits will be to tubular flowers with a 95% credible interval from [30% - 43%], and 41% of Butterfly/Moth visits will be to tubular flowers with a 95% credible interval from [28% - 54%].

Biologically this means that 66% of Fly visits will be to dish flowers with a 95% credible interval from [45% - 75%], 49% of Bee visits will be to dish flowers with a 95% credible interval from [33% - 46%], and 40% of Butterfly/Moth visits will be to tubular flowers with a 95% credible interval from [25% - 55%].

Graphs

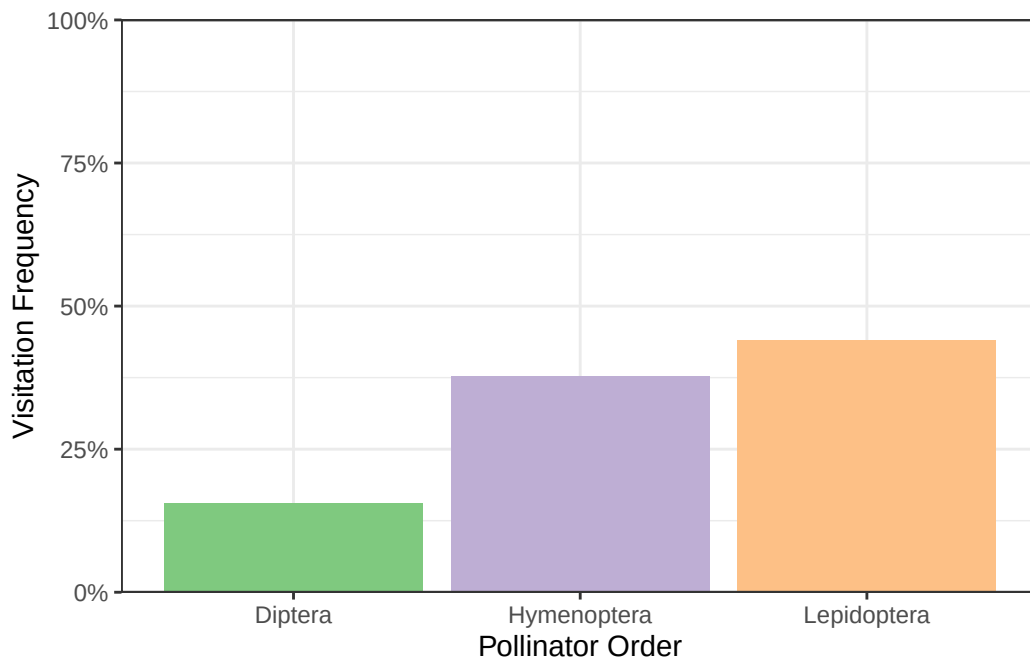
```
# create graph for visitation frequency on tubular flowers
TubeGraph <- ggplot(Tube_Fl_Order, aes(PollinatorOrder, y = prop)) +
  # make it a bar plot and set fill = PollinatorOrder for color later
```

```

geom_bar(stat = "identity", mapping = aes(fill = PollinatorOrder)) +
# Change x and y axis
labs(x = "Pollinator Order",
     y = "Visitation Frequency") +
# get rid of gap at y = 0 and make y-axis percentage
scale_y_continuous(expand = c(0, 0), labels = scales::percent, limits = c(NA, 1)) +
# change theme
theme_bw() +
# remove legend
theme(legend.position = "none") +
# change colors of bars to look nicer
scale_fill_brewer(palette = "Accent")

# print out graph
TubeGraph

```



```

# create graph for visitation frequency on dish flowers
DishGraph <- ggplot(Dish_Fl_Order, aes(PollinatorOrder, y = prop)) +
# make it a bar plot and set fill = PollinatorOrder for color later
geom_bar(stat = "identity", mapping = aes(fill = PollinatorOrder)) +
# Change x and y axis
labs(x = "Pollinator Order",

```

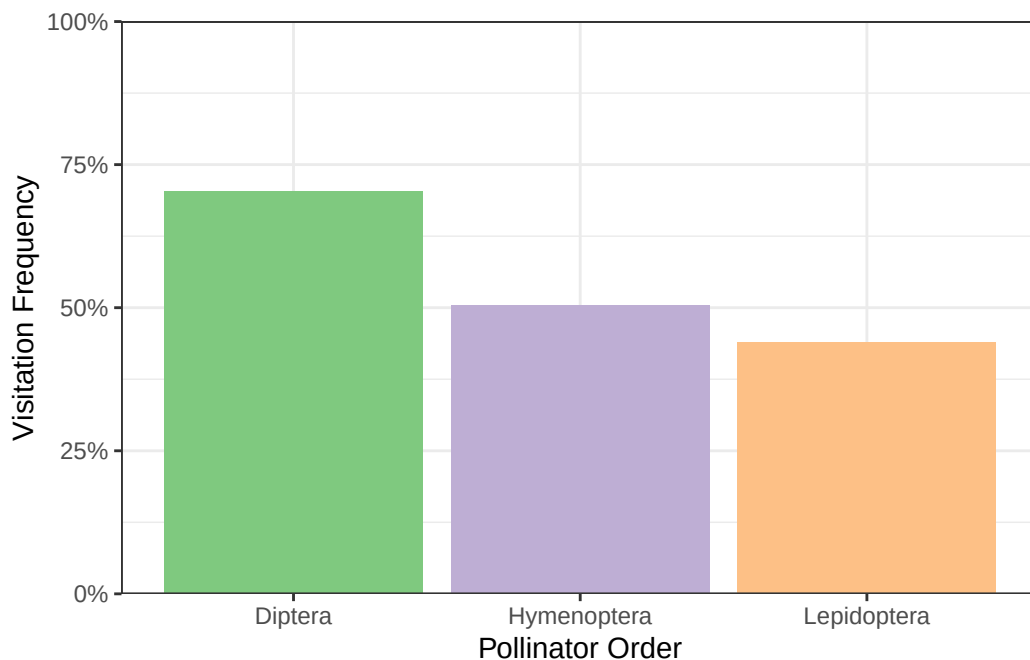


```

    y = "Visitation Frequency") +
  # get rid of gap at y = 0 and make y-axis percentage
  scale_y_continuous(expand = c(0, 0), labels = scales::percent, limits = c(NA, 1)) +
  # change theme
  theme_bw() +
  # remove legend
  theme(legend.position = "none") +
  # change colors of bars to look nicer
  scale_fill_brewer(palette = "Accent")

# print out graph
DishGraph

```



Based on these 2 figures you can see how Diptera or Flies have a strong tendency to visit flowers that are dish shaped. They also have very little tendency to visit tubular flowers when compared to the other orders examined. With Butterflies/moths we can see that they have the exact same visitation frequency to dish and tube flowers; this could indicate a lack of preference, however our data for this order is a bit limited so perhaps with more data a pattern could emerge. With Hymenoptera or Bees we can see that they tend to visit dish flowers a bit more frequently than tubular flowers; this could be due to their mouthparts being more adapted for flowers with less of a distance to nectar (similar to flies). Perhaps it is different for different families of bees, and further analysis could help us to understand this.